Hands-on Lab: Getting Started with GitHub

Effort: 20 min

In this lab, you will get started with GitHub by creating a GitHub account and project and adding a file to it using its Web interface.

Objectives

After completing this lab, you will be able to:

0. Describe GitHub
1. Create a GitHub account
2. Add a Project / Repo
3. Edit / Create a file
4. Upload a file & Commit

GitHub Overview

First, let us introduce to GitHub. GitHub in simple words is a collection of folders and files. It is a Git repository hosting service, but it adds many of its own features. While Git is a command-line tool and a server needs to be hosted and maintained via command line as well, GitHub provides this Git server for you and a Web-based graphical interface. It also provides access control and several collaboration features, such as wikis and basic task management tools for every project. GitHub provides cloud storage for source code, supports all popular programming languages, and streamlines the iteration process. GitHub includes a free plan for individual developers and for hosting open source projects.

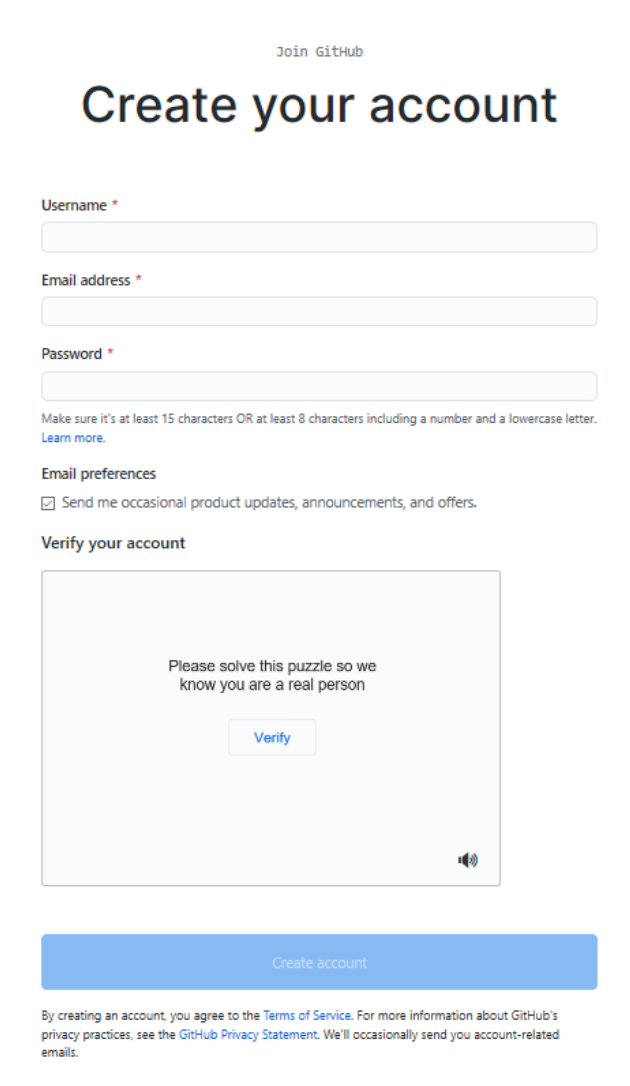
Exercise 1: Creating a GitHub Account

Please follow the steps given below to create an account in GitHub:

Step 1: Create an account: https://github.com/join

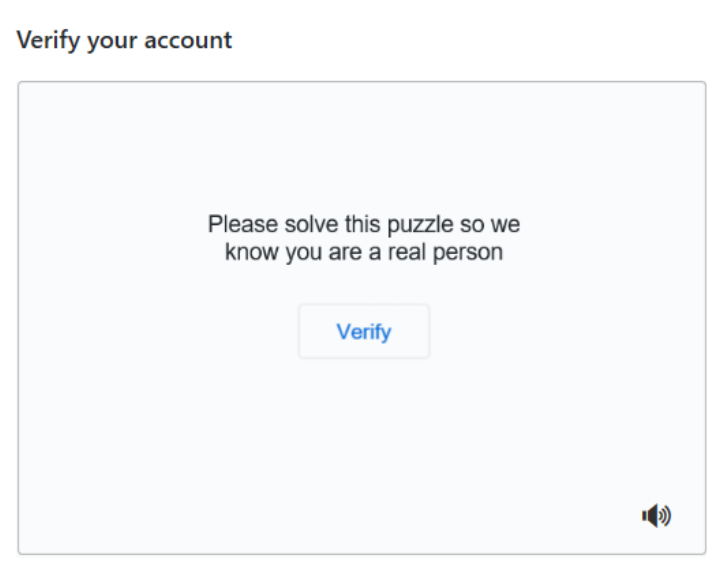
NOTE: If you already have a GitHub account, you can skip this step and simply login to your account.

Step 2: Provide the necessary details to create an account as shown below:

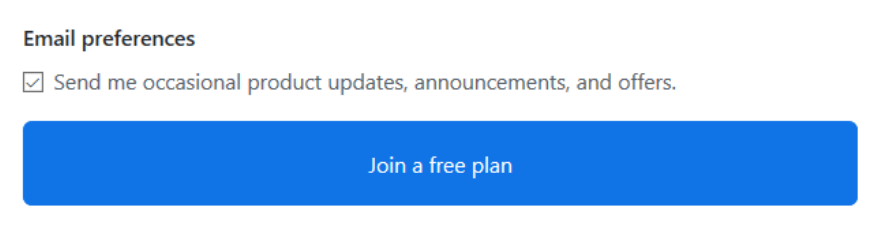

Join GitHub
Create your account
Username *
Email address *
Password *
Make sure it's at least 15 characters OR at least 8 characters including a number and a lowercase letter. Learn more.
Email preferences
Send me occasional product updates, announcements, and offers.
Verify your account
Please solve this puzzle so we know you are a real person
Verify
Create account
By creating an account, you agree to the Terms of Service. For more information about GitHub's privacy practices, see the GitHub Privacy Statement. We'll occasionally send you account-related emails.

and click `Create account`.

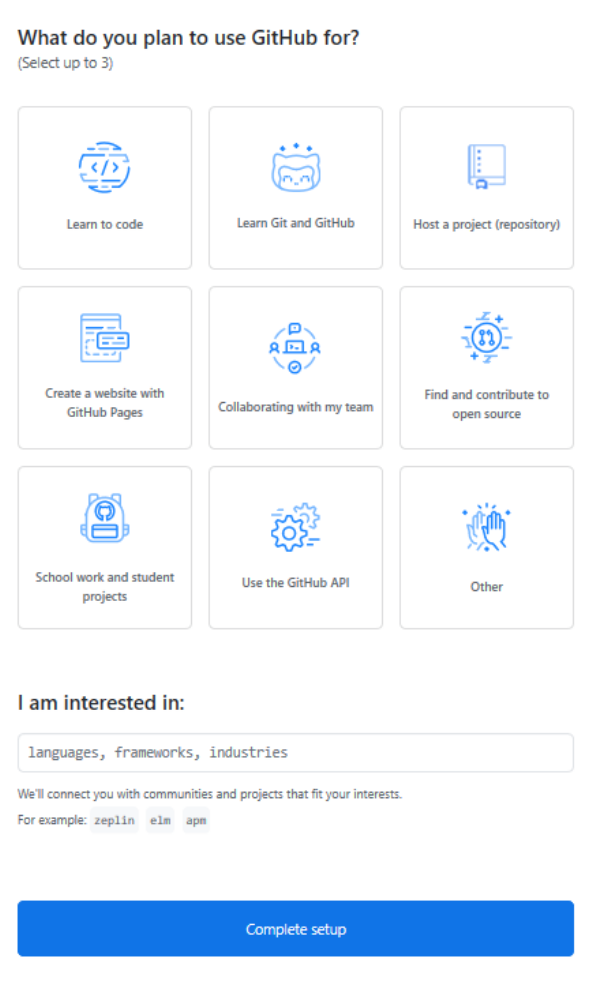
Step 3: Click `Verify` to verify the account and click `Done`


Verify your account
Please solve this puzzle so we know you are a real person
Verify

Step 4: After verification, click `Join a Free Plan`

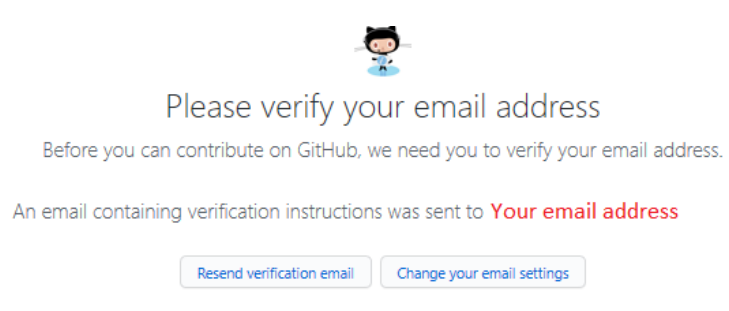

Email preferences
Send me occasional product updates, announcements, and offers.
Join a free plan

Step 5: Select the details as shown below and click `Complete Setup`

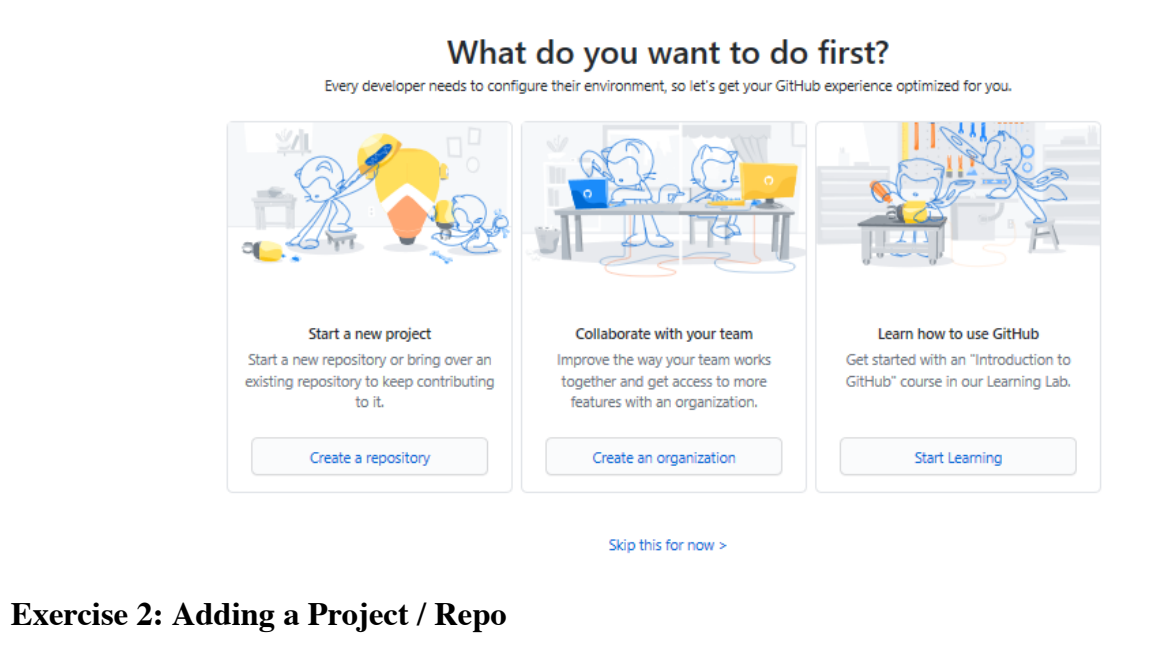

What do you plan to use GitHub for?
(Select up to 3)
Learn to code
Learn Git and GitHub
Host a project (repository)
Create a website with GitHub Pages
Collaborating with my team
Find and contribute to open source
School work and student projects
Use the GitHub API
Other
I am interested in:
languages, frameworks, industries
We'll connect you with communities and projects that fit your interests.
For example: zeplin elm npm
Complete setup

Step 6: Go to your email, find the verification email from GitHub, and click on the link/button in that email to verify your email.

NOTE: If you do not receive verification email, click `Resend verification email`.

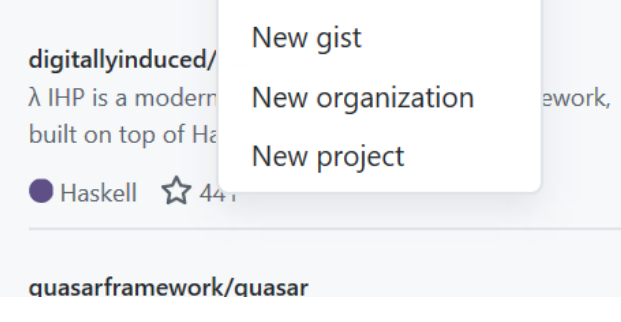

Please verify your email address
Before you can contribute on GitHub, we need you to verify your email address.
An email containing verification instructions was sent to Your email address
Resend verification email
Change your email settings

Email is verified

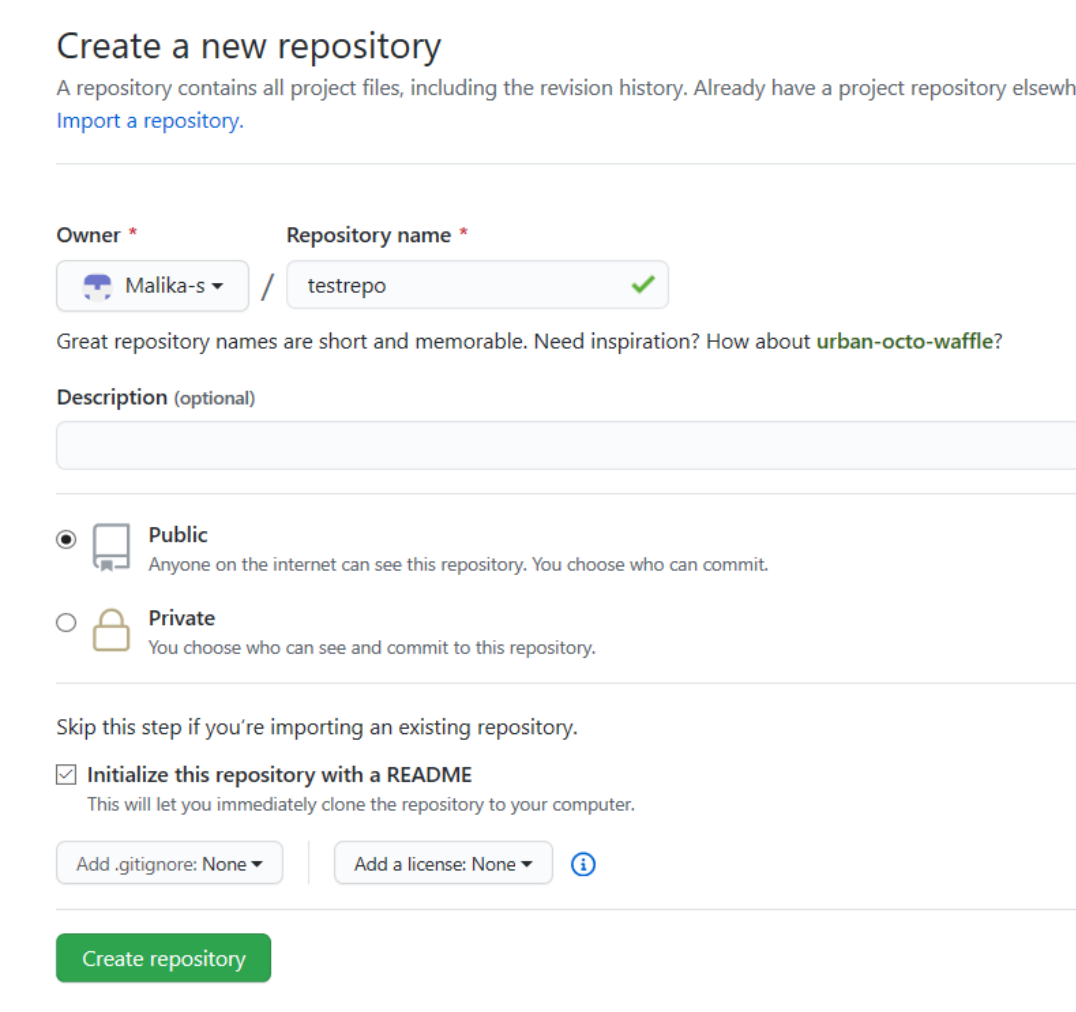

Your email was verified.
What do you want to do first?
Every developer needs to configure their environment, so let's get your GitHub experience optimized for you.
Start a new project
Start a new repository or bring over an existing repository to keep contributing to it.
Create a repository
Collaborate with your team
Improve the way your team works together and get access to more features with an organization.
Create an organization
Learn how to use GitHub
Get started with an "Introduction to GitHub" course in our Learning Lab.
Start Learning
Skip this for now >

Exercise 2: Adding a Project / Repo

Step 1: Click on the + symbol and click `New repository`.


New repository
Import repository
New gist
New organization
New project
Explore reposi
digitallyinduced/
λ IHP is a modern
built on top of Ha
ework,
Haskell
quasarframework/quasar

Step 2: Provide a repository a name and initialize with the empty `README.md` file.


Create a new repository
A repository contains all project files, including the revision history. Already have a project repository elsewhere?
Import a repository.
Owner *
Repository name *
Malika-s
testrepo
Great repository names are short and memorable. Need inspiration? How about urban-octo-waffle?
Description (optional)
Public
Anyone on the internet can see this repository. You choose who can commit.
Private
You choose who can see and commit to this repository.
Skip this step if you're importing an existing repository.
Initialize this repository with a README
This will let you immediately clone the repository to your computer.
Add .gitignore: None
Add a license: None
Create repository

and click `Create repository`.

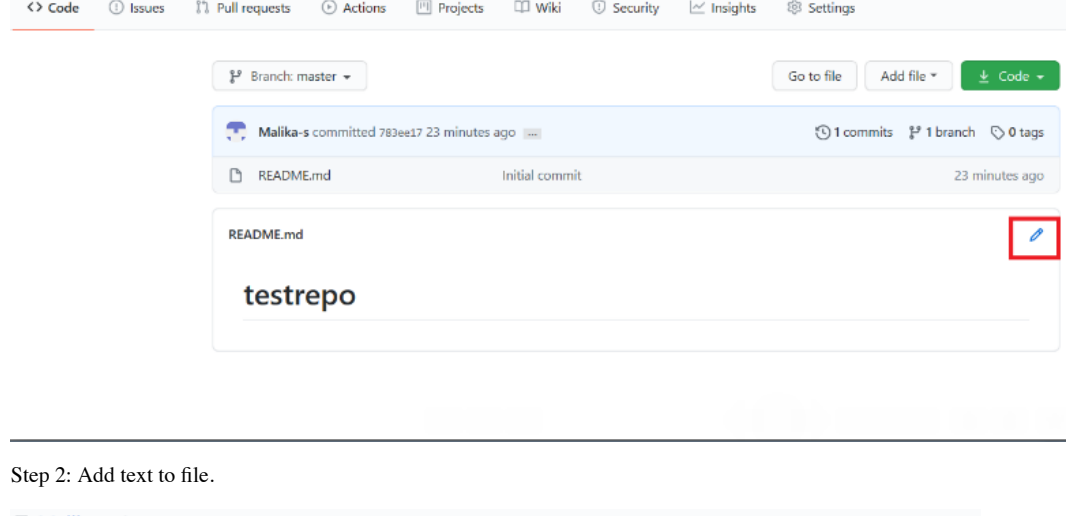
Now, you will be redirected to the repository you have created.

Let's start editing the repository.

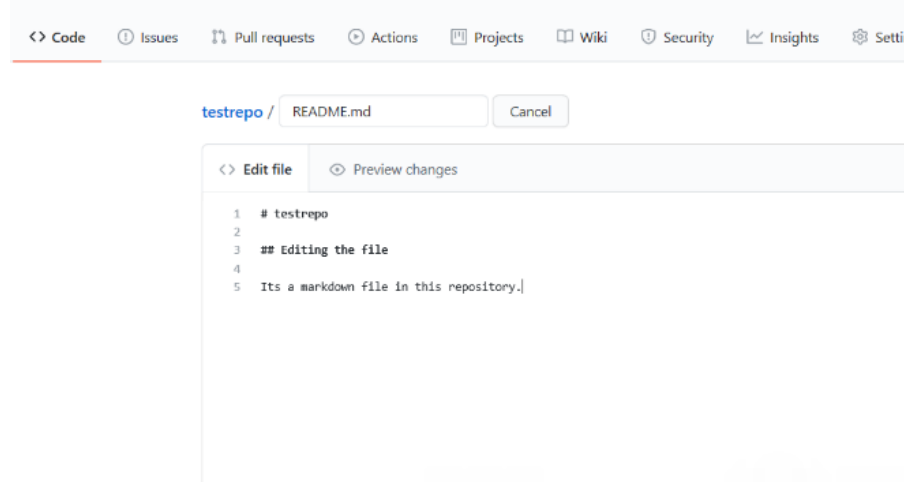
Exercise 3: Create / edit a file

Exercise 3a: Edit a file

Step 1: Once the repository is created, the root folder of your repository is listed by default and it has just one file `ReadMe.md`. Click on the pencil icon to edit the file.


Malika-s / testrepo
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Branch: master
Go to file
Add file
Code
Malika-s committed 783ee17 23 minutes ago
1 commits
1 branch
0 tags
README.md
Initial commit
23 minutes ago
README.md
testrepo

Step 2: Add text to file.

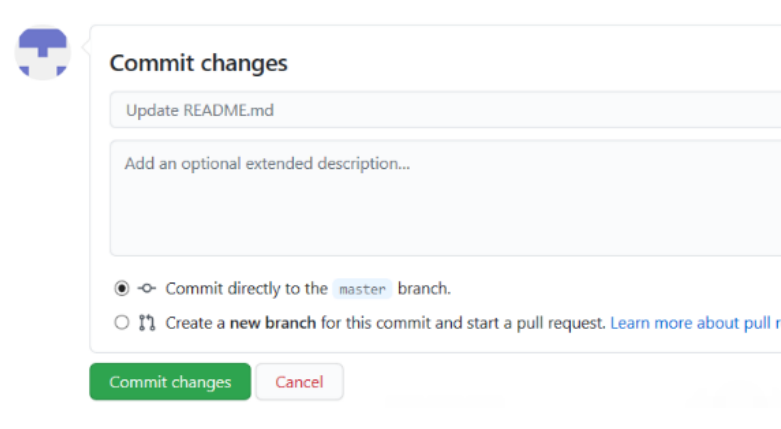

Malika-s / testrepo
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
testrepo / README.md
Cancel
Edit file
Preview changes

```
# testrepo

## Editing the file

Its a markdown file in this repository.
```

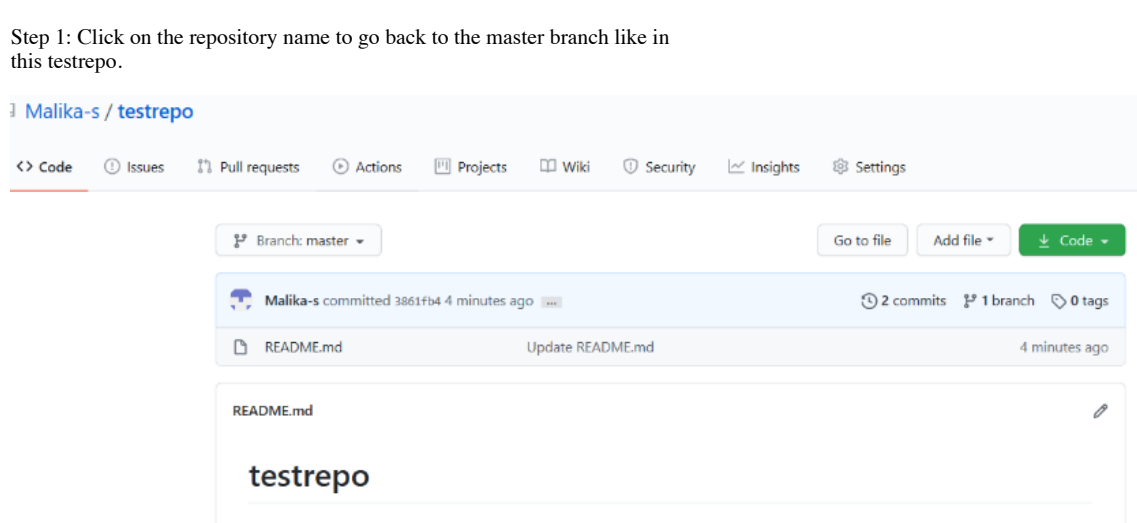
Step 3: Scroll down the page after adding the text and click `Commit Changes`.


Commit changes
Update README.md
Add an optional extended description...
Commit directly to the master branch.
Create a new branch for this commit and start a pull request. Learn more about pull requests.
Commit changes
Cancel

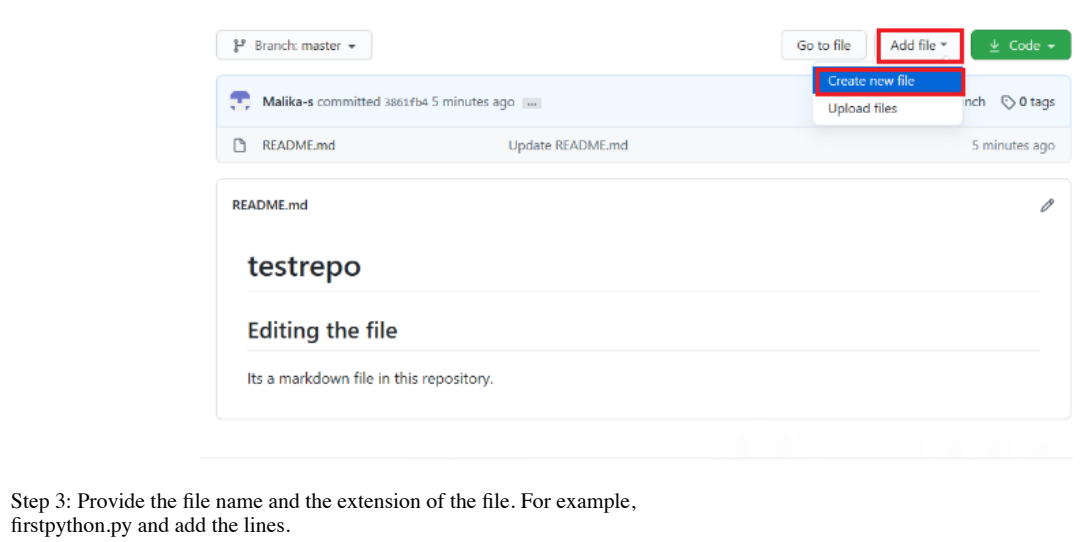
Now, check your file is edited with the new text.

Exercise 3b: Create a new file

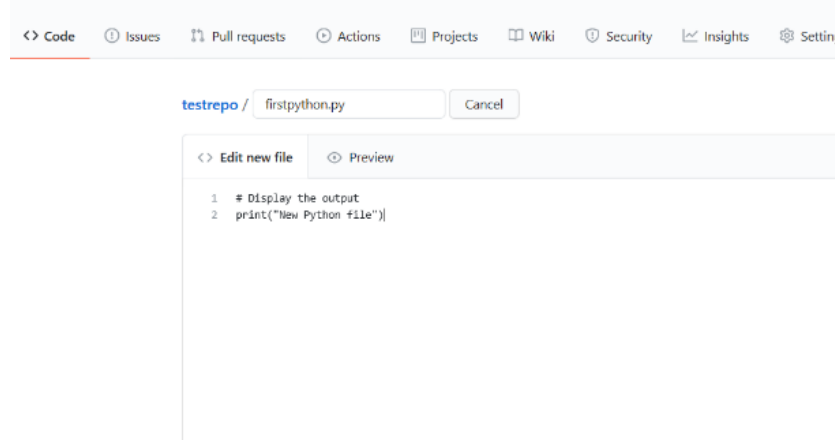
Step 1: Click on the repository name to go back to the master branch like in this testrepo.


Malika-s / testrepo
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Branch: master
Go to file
Add file
Code
Malika-s committed 3861fb4 4 minutes ago
2 commits
1 branch
0 tags
README.md
Update README.md
4 minutes ago
README.md
testrepo
Editing the file
Its a markdown file in this repository.

Step 2: Click `Add file` and select `Create New file` to create a file in the repository.

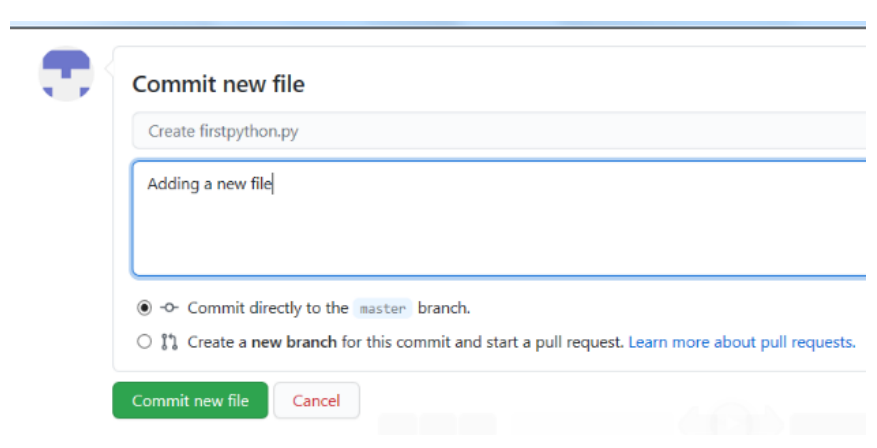

Malika-s / testrepo
Branch: master
Go to file
Add file
Code
Create new file
Upload files
Malika-s committed 3861fb4 5 minutes ago
0 tags
README.md
Update README.md
5 minutes ago
README.md
testrepo
Editing the file
Its a markdown file in this repository.

Step 3: Provide the file name and the extension of the file. For example, firstpython.py and add the lines.


Malika-s / testrepo
testrepo / firstpython.py
Cancel
Edit new file
Preview

```
# Display the output
print("New Python file")
```

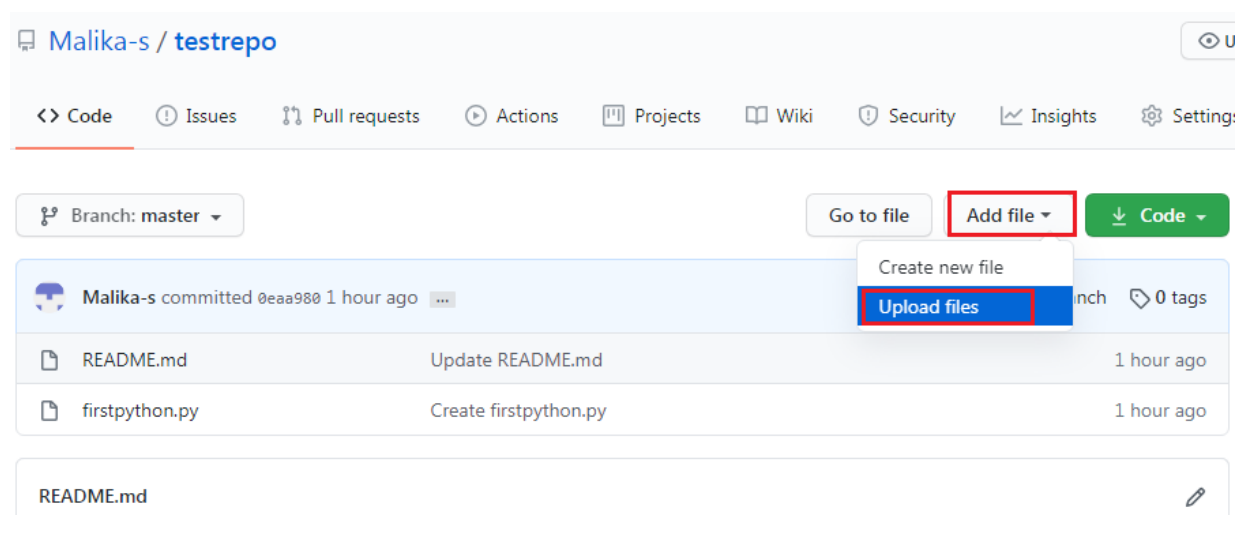
Step 4: Scroll down the page after adding the text. `Add description` of the file (optional) and click `Commit` new file.


Commit new file
Create firstpython.py
Adding a new file
Commit directly to the master branch.
Create a new branch for this commit and start a pull request. Learn more about pull requests.
Commit new file
Cancel

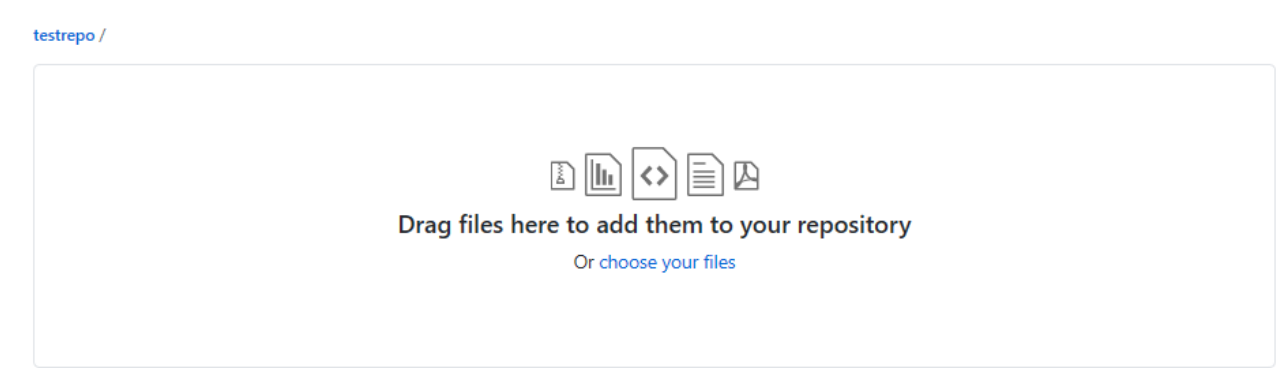
Step 5: Your file is now added to your repository and the repository listing shows when the file was added/changed.

Exercise 4: Upload a file & Commit

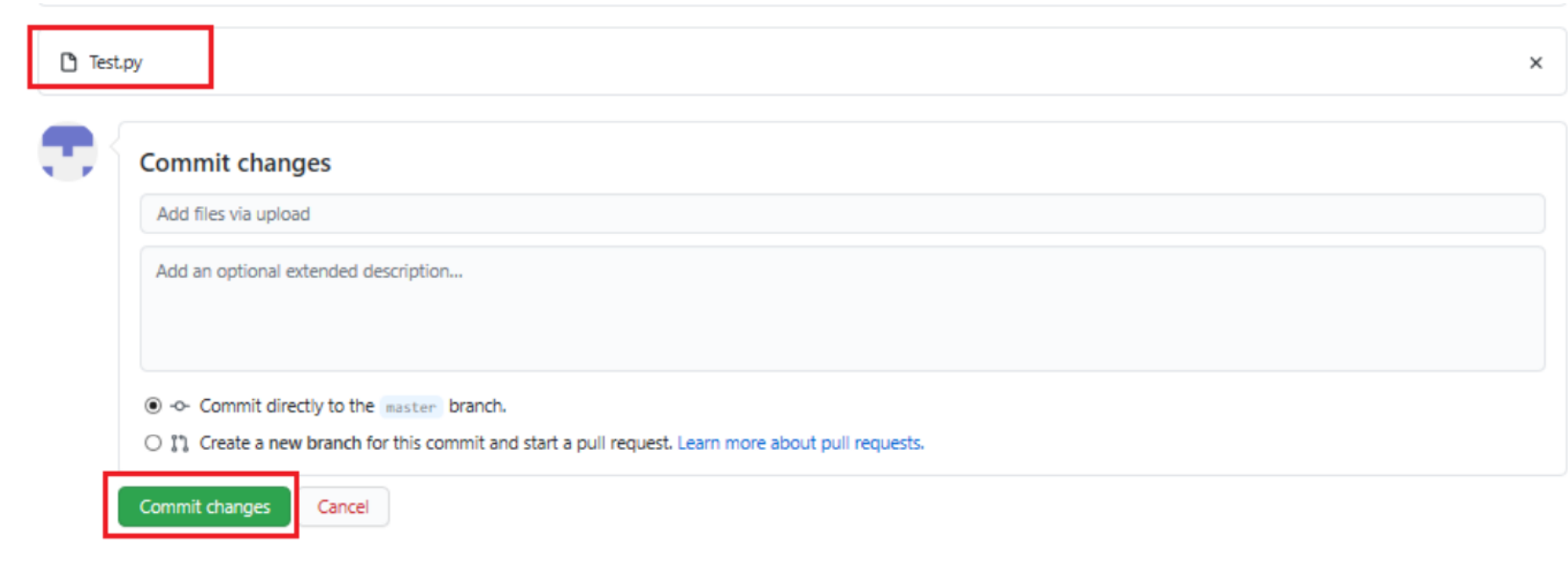
Step 1: Click `Add file` and select `Upload files` to upload a file (Upload any .txt,.ipynb, .png file) in the repository from the local computer.


Malika-s / testrepo
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Branch: master
Go to file
Add file
Code
Create new file
Upload files
Malika-s committed 0eaa980 1 hour ago
0 tags
README.md
Update README.md
1 hour ago
firstpython.py
Create firstpython.py
1 hour ago
README.md

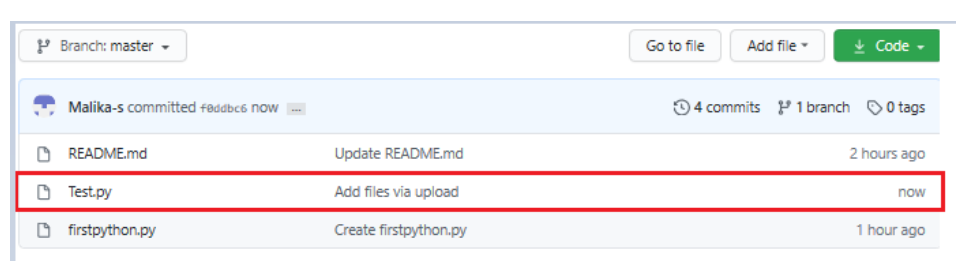
Step 2: Click on choose your files and choose any files from your computer.


testrepo /
Drag files here to add them to your repository
Or choose your files

Step 3: Once the file finishes uploading, click on `Commit` Changes


Test.py
Commit changes
Add files via upload
Add an optional extended description...
Commit directly to the master branch.
Create a new branch for this commit and start a pull request. Learn more about pull requests.
Commit changes
Cancel

Step 4: Now, your file is uploaded in the repository.


Branch: master
Go to file
Add file
Code
Malika-s committed f8ddbc6 now
4 commits
1 branch
0 tags
README.md
Update README.md
2 hours ago
Test.py
Add files via upload
now
firstpython.py
Create firstpython.py
1 hour ago

Summary

In this document, you have learned how to create a new repository, adding a new file, editing a file, and uploading a file in a repository and commit the changes.

Author(s)

Romeo Kienzler

Malika Singla

Other Contributor(s)

Rav Ahuja

Skills Network

IBM